Q11. Nmap service and version detection

Original question: Perform service and version detection on an authorized lab target. Identify open ports and associated services, interpret the results, and document potential security concerns.

Aim

Identify service implementations and reported versions on an authorized lab target, with careful interpretation of uncertainty.

Requirements

Nmap, one assigned IP, permission for service probes and the approved TCP port set. Unlike Q4's port-discovery emphasis, this practical focuses on application responses. Reference: [R10].

Procedure

Step 1 - Record context and discover ports

```
mkdir -p "$HOME/EHF_Exam/Q11"
cd "$HOME/EHF_Exam/Q11"
TARGET='<LAB_TARGET_IP>'
date -Is | tee collected_at.txt
nmap --version > nmap_version.txt
nmap -n -sT --reason -oN port_discovery.txt "$TARGET"
cat port_discovery.txt
```

Use only the assigned host. The default TCP scan covers the most common 1,000 ports, so explicitly record that scope. If the examiner provides a smaller approved list, use `-p` with that list instead.

Step 2 - Probe services on discovered open TCP ports

```
PORTS='<OBSERVED_OPEN_TCP_PORTS>'
nmap -n -sT -sV --reason -p "$PORTS" \
  -oA service_versions "$TARGET"
cat service_versions.nmap
ls -l service_versions.*
```

Replace PORTS with actual comma-separated open ports. `-sV` sends application probes to identify service, product and version; `-oA` saves normal, XML and grepable formats using the given prefix. No root is needed for this TCP connect example. Use `--version-light` only when fewer probes are desired; it may identify less than the default version scan. If identification remains uncertain, retain that uncertainty rather than increasing probes without a reason.

Step 3 - Evaluate the evidence

Read PORT, STATE, SERVICE and VERSION together. Copy product/version exactly, including question marks or uncertainty. A `tcpwrapped` result suggests a connection was closed early; it is not the product name. Where a banner is absent or generic, record version unknown. Nmap can recognize TLS-wrapped services, but encryption and proxies can obscure the application. Do not use `-A` just to get versions: it enables additional features beyond this question.

Expected output - illustrative

```
PORT    STATE SERVICE VERSION
22/tcp  open  ssh     OpenSSH 9.x (protocol 2.0)
80/tcp  open  http    nginx
```

This synthetic table shows the format, not an exact Nmap signature. Do not substitute it for your scan. A version string may come from a banner and may not reflect backported security patches.

Interpretation, observation and result

For each port record: service, product/version, evidence confidence, possible concern and proposed verification. A service on a nonstandard port should be identified by its response, not the conventional port label. Review exposed remote access, plaintext services and potentially unsupported software with the host's role and vendor patch inventory. A version match alone does not establish a CVE is exploitable; confirm configuration and vendor advisories through a separately approved assessment. The result is a saved, qualified inventory of detected services/versions, not a vulnerability report.

Common errors

- No VERSION column detail: banners may be hidden or the protocol unsupported; record uncertainty.
- Host discovery blocked: confirm the assigned host is up, then use -Pn if appropriate within scope.
- TLS name-based application not identified: an IP scan may not represent the named virtual host; do not invent a product.

Viva questions

- How is -sV different from a SERVICE port label? It uses protocol probes and responses for identification.
- What does -oA save? Normal, XML and grepable scan files.
- Does -sV detect every vulnerable version? No; fingerprinting and patch-state inference have limits.
- Why avoid -A for this question? It adds OS detection, scripts and traceroute beyond version detection.